

The Rank-21 Program: Reproduction, Soundness, Exhaustive Negatives, and a Certified-Search Architecture for $\langle 3, 3, 3 \rangle$ over $\mathbb{F}_2$

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Abstract

We report on a program aimed at improving the lower bound for the rank of the 3×3 matrix multiplication tensor over $\mathbb{F}_2$, currently $20 \leq R_{\mathbb{F}_2}(\langle 3, 3, 3 \rangle) \leq 23$ (Wang’s automated lower bound; Laderman’s scheme reduced mod 2). The contributions are: (i) an independent reproduction and strengthening of Wang’s certified DP-over-restrictions bound, including budget-scaled lifts of his orbit lattice; (ii) a sound extension of his framework to *mixed-side* substitution, including an activity guard whose necessity we demonstrate by measurement, and a family of exhaustive negative results showing that substitution with flattening-class leaves cannot exceed the lattice values; (iii) an exact determination of our substitution game’s value (18) under successively stronger rule sets, with machine-checked certificates; (iv) a certified-search (CDCL(T)-style) architecture over scheme space whose propagator and symmetry components each deliver their theoretical strength (measured ablations of $746\times$ and $632,384\times$), reaching an independent, exhaustively-searched $R_{\mathbb{F}_2} > 15$ in 3.25 hours; and (v) an empirical structure theorem for the adversary’s dangerous moves in the substitution game — *the Rank Law* — identifying them with rank-one covectors, which yields a concrete, well-posed lemma whose proof would extend the certified ladder at $\sim 15\times$ reduced cost. We close with a convergence thesis: three structurally different attacks terminate at the same missing object, a cheap rank lower-bound oracle of strength 15–19 for arbitrary tensors, and we locate the genuine frontier at scheme size 20, where every known bound is structurally one short of what novelty requires.

1 Setting and known bounds

Let $\mathcal{T} = \langle 3, 3, 3 \rangle \in \mathbb{F}_2^9 \otimes \mathbb{F}_2^9 \otimes \mathbb{F}_2^9$ denote the matrix multiplication tensor in the trace convention $\mathcal{T} = \sum_{i,j,k} x_{ij} \otimes y_{jk} \otimes z_{ki}$, a 729-bit object with 27 nonzero entries. Its rank $R_{\mathbb{F}_2}(\mathcal{T})$ is the minimum number of products $\alpha \otimes \beta \otimes \gamma$ (rank-one tensors) whose XOR is $\mathcal{T}$; schemes of r products are exactly solutions of the Brent equations with r multiplications. Known bounds: $R_{\mathbb{F}_2}(\mathcal{T}) \geq 20$ (Wang, arXiv:2603.07280, with a machine-verifiable certificate of ~ 32 MiB), and $R_{\mathbb{F}_2}(\mathcal{T}) \leq 23$ (Laderman’s scheme reduced mod 2; independently, our 55-addition rank-23 scheme). Bläser’s ≥ 19 holds over every field. The program’s goal is 21.

Throughout, *flattening* bounds are matrix ranks of the three unfoldings (≤ 9 here); K_p denotes the Koszul flattening with wedge parameter p (division bound $\lceil \text{rank} / \binom{8}{p} \rceil$); and the *substitution lemma* is used in the form: for an active direction (one the tensor depends on), $R(T) \geq 1 + \min_{\lambda} R(T|_{\lambda})$ where λ ranges over all foldings of that direction.

A structural datum that organizes everything below: rank methods — bounds expressible as matrix ranks of linearly-constructed flattenings, including Koszul, Young and Strassen’s equa-

tions — are barrier-limited near $2n^2 = 18$ for matmul-type tensors. Our measurements respect this barrier with uncanny precision: flattenings reach 9, Strassen’s commutator bound 12–13, Koszul 14–16, our exhaustive game with such leaves exactly 18. Wang’s 20 lives beyond the barrier because substitution-DP is *recursive amplification*, not a rank method. This is the frame in which “new leaf mathematics” must be read (§6, §7).

2 Reproducing and strengthening the lattice DP

We reproduced Wang’s result locally from his repository (verifier included), then strengthened his certificate by re-running his backtracking technique at $30\times$ the shipped step budget over the whole restriction lattice. The wave lifted every layer — e.g. the fourteen codimension-2 orbits from $\{17\times 1, 18\times 13\}$ to $\{18\times 6, 19\times 8\}$ — but the root held at 20: a diminishing-returns profile, quantified. (A root of 21 requires, among more, all fourteen codim-2 orbits at ≥ 19 ; six remain at 18 and resisted 3×10^9 steps each.) Two engineering contributions accompany the reproduction: a diagnosis and fix of an allocator-retention memory ratchet in long runs (inline DFS-cache keys; heap pressure relief), and per-orbit byte-level certificate merging. All lifted certificates re-verify under his verifier.

3 Mixed-side substitution: soundness and negatives

Wang’s framework restricts only the *A*-side. Motivated by our game measurements (*A*-only value 16 vs. mixed value 18; §4), we implemented the *fifth technique*: chains mixing *A*- and *B*-side substitutions inside his backtracking DFS, with a backward-compatible proof format (per-record leaf kinds, per-node side bits) and a mirrored verifier.

The activity guard. The naive mixed-leaf bound (chain length plus flattening of the doubly-restricted tensor) is *unsound*: pure-*A* chains get activity for free (*A*-flattening injectivity survives *A*-restriction), but under double-sided restriction a substitution can kill a direction the tensor no longer depends on, inflating the bound by +1 per inactive step. The sound credit is rank-based: $k_B = \dim B - \dim(W + D)$, where W is the kernel of the *B*-rows and D the dead *B*-subspace of the *A*-restricted tensor. We validated the formula on engineered deactivations (independent Python implementation and a C++ probe), then measured it in production: per-orbit docked credit ranged from 0 to 8.6×10^9 — the guard is not vacuous, and the unguarded rule’s 99.99% “rescue rates” were partly phantom credits.

Exhaustive negatives. With soundness in place, equal-budget experiments returned a clean null: guarded mixed play lifted no codim-2 orbit at 3×10^8 steps (nor did *A*-only). Escalating down-lattice where exhaustion is affordable produced eight decisive verdicts, *all FAILS*: at codim-4 and codim-3, mixed substitution with table leaves cannot lift Wang’s values by one, exhaustively. A three-sided variant (*C*-moves) inflates the game ($\geq 6.6\times$ nodes on refutations that were 16–40 minutes under two sides) with no observed proving power. The binding constraint is identified and measured: once play leaves the pure-*A* lattice, every available leaf collapses to flattening class (≈ 14 –16), and the adversary grinds targets down against them.

4 The exhaustive substitution game

Independently of Wang’s framework, we built an exact game engine over (U_A, U_B, U_C) restriction states with sandwich-group canonicalization, stabilizer-reduced adversary fanouts, and memoized decision ladders. Its verdicts, each with machine-checked certificates replayed by an independent Python checker (including isomorphism edges): game value 17, then 18 under support/spanning refinements and rank-profile case splits; three exhaustive FAILS at 19 under every rule set tried. A methodological lesson is recorded with the mathematics: an early version “proved” $R_{\mathbb{F}_2}(\langle 2, 2, 2 \rangle) \geq 12$ (true value 7) because kill moves scored on the zero tensor *and the checker re-encoded the same rule*. Certificate checkers validate transcription, not semantics; every new proof rule since has been gated on known exact values first.

5 Certified scheme search (CDCL(T) over products)

The constructive dual: prove $R_{\mathbb{F}_2} > r$ by exhausting r -product schemes, with theory propagators supplying the reasoning that resolution lacks. Decisions are whole products; the residual (tensor XOR chosen products) carries native parity; propagators prune by residual rank bounds; symmetry enters as constructive orbit enumeration.

Gates at 2×2 . The stage-1 prototype reproduces both certified values: $r = 6$ UNSAT (matching our hydra_satsuma DRAT-verified result) and $r = 7$ SAT with an independently verified Strassen-class witness. The measured ablation: one ply of sound substitution reasoning cut the $r = 6$ tree from 3.22×10^9 to 4.3×10^6 nodes (746 $\times$).

Scaling to 3×3 . Lazy enumeration over the $511^3 \approx 1.3 \times 10^8$ products; triple-layout residuals making every flattening nine contiguous 81-bit extracts; constructive first-product canonicalization under the sandwich group ($133,432,831 \rightarrow 211$ orbit representatives; the level-1 frontier *is* the orbit count); level-2 stabilizer minimality (median 16 $\times$); a Strassen-commutator propagator (strength 12–13 at 10–30 μ s, validated by a constructed-rank gate: tensors built as XOR of k products have rank $\leq k$ by construction); Koszul at the shallow levels. The ladder:

rung	verdict	nodes	wall
$r = 9$	UNSAT (flatten)	783,363 $\rightarrow$ 211	0.1 s
$r = 10$	UNSAT (probe)	211	0.1 s
$r = 11$ –14	UNSAT (Koszul at level 1)	211 each	0.1 s each
$r = 15$	UNSAT (killer-directed deep probe)	211	3.25 h
$r = 16$	out of reach	—	≥ 70 core-h on one root, unresolved

The $r = 15$ rung is the architecture’s current independent reach, $R_{\mathbb{F}_2} > 15$: two rungs past the Koszul ceiling, taken by *lazy adversary-directed recursion* — at each node a cheap-floor chain (flatten $\rightarrow$ Strassen $\rightarrow$ Koszul) settles most folds, and only the *killers* (Koszul-dropping folds) recurse, with targets falling into Strassen range within two levels. Fixed-sweep plies, by contrast, were measured intractable (16.7 hours resolving fewer than 8 of 211 roots). Engineering lessons are recorded honestly: Method-of-Four-Russians elimination, equality-gated, was *benched out* (dense wins do not transfer to the sparse wedge shape); a leading-bit rewrite of the 9×81 rank

kernel gave a universal $2.1\times$; five unsound artifacts across the program (an α -canonical cut, a $\lambda=0$ -only probe, side-slot pollution, activity-free mixed leaves, a row-reversed matrix inverse) were all caught by known-value or constructed-value gates before contaminating any claim.

6 The Rank Law

Instrumenting the $r = 15$ probes produced a dataset of every Koszul-dropping fold at the 211 roots (10,338 killers; the probe’s pivot is always the cell $(2, 2)$; every drop is $14 \rightarrow 13$ exactly). Writing the folding covector as $\varphi = e_{(2,2)} + \lambda$ and viewing it as a 3×3 matrix over $\mathbb{F}_2$:

Observation 1 (Rank Law; 211 roots $\times$ 3 sides = 633 pairs).

1. $\text{rank}(\varphi) = 1 \implies$ *killer, with zero exceptions. These are the $4 \times 4 = 16$ rectangle indicators through the pivot cell — exactly the observed sixteen killers per pair.*
2. $\text{rank}(\varphi) = 3 \implies$ *non-killer, with zero exceptions.*
3. $\text{rank}(\varphi) = 2$: *non-killer at 595 of 633 pairs; killer only at 38 exceptional pairs concentrated on 30 roots (210 extra folds).*

The reading: the adversary’s dangerous moves are *product-shaped* — rank-one covectors aligned with the tensor’s own rank-one structure, the same objects our game’s rank-profile splits singled out. The law immediately poses a well-defined lemma:

Conjecture 1 (case-split lemma). *On the level-1 residuals (matmul minus one product, restricted as in the probe), folding by a covector of matrix rank 3 preserves $K_4 \geq 14$.*

Proven — and extended across the rank-2 boundary, whose 38 exceptions appear tied to low-rank structure of the root representatives — the lemma removes 168–240 of every 256 probe evaluations, repricing the $r = 16$ rung from ≥ 1.5 core-years to days. Independent of that payoff, the law is the first structural characterization of the substitution game’s critical directions that we are aware of, and the killer set’s algebraic description (points of an affine determinantal pencil over $\mathbb{F}_2^8$) suggests it is provable.

7 Convergence, the frontier at $r = 20$, and outlook

Three structurally different programs — mixed-side substitution inside Wang’s DP, the exhaustive game, and certified scheme search — terminate at the same gap: a cheap ($\lesssim$ ms) rank lower-bound oracle of strength 15–19 for *arbitrary* tensors (lattice-restricted tensors have Wang’s table; residuals do not). The barrier section explains why the gap is stubborn: that strength range spans the rank-method ceiling, so the oracle must itself be amplification-shaped.

For the program’s actual goal, the frontier is sharper than any oracle question. Subadditivity plus Wang’s certificate gives $\text{rank}(T \oplus m_1 \oplus \cdots \oplus m_k) \geq 20 - k$ for free, which trivializes every rung $r \leq 19$; proving $R_{\mathbb{F}_2} > 20$ requires certifying $21 - k$ on *scheme-consistent* prefixes — beating subadditivity by exactly one, which at $k = 0$ is the theorem itself. The $+1$ must come from knowledge about minimal decompositions of $\mathcal{T}$ specifically. Two routes: (a) prove and iterate the Rank Law (killer varieties as case-split rules — the game’s critical directions may thin as prefixes accumulate); (b) decomposition-structure theorems in the tradition of de Groote’s uniqueness and Burichenko’s symmetries, mined from the known scheme corpora and imported as WLOG

prunes. On the constructive side, the corridor experiment established that prefix descent has no usable guidance signal (flattening is capped and constant; true schemes' intermediates are low-rank, not low-popcount), so the rank-22 search correctly factors as SLS near-misses (native-ANF Brent engine) finished or refuted by the exact endgame engine, which refutes junk prefixes in milliseconds.

All measurements, certificates, logs and the killer dataset are in the repository; every number above is reproducible from committed artifacts.

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Acknowledgment of method. The program's productivity traces to two disciplines applied without exception: gate every new rule on known exact values (five unsound artifacts caught), and distrust suspicious nulls until the treatment is proven applied (three vacuous experiments unmasked). The instruments built for soundness — the activity-deficit counters, the killer dump — produced the mathematical discoveries.